Allocating Specification Shifts with Correlated Controls

One-at-a-time, conditional leave-one-out, and Shapley diagnostics

Online appendix for the IVB paper

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Contents

1 Purpose and scope

When candidate controls are correlated, the joint specification shift is unique but its control-level allocation is not. Let $\beta(S)$ denote the coefficient on the focal treatment in the linear model containing the common regressors W and the subset S of candidate controls, all estimated on the same observations. Define

$$\Delta(S) = \beta(S) - \beta(\emptyset), \qquad \Delta_{\mathcal{Z}} = \beta(\mathcal{Z}) - \beta(\emptyset).$$

This appendix compares three descriptive diagnostics. Each is a contrast between nested linear projections; none establishes causal bias relative to the contemporaneous treatment effect (CET). In a linear TSCS application, W can include unit and time fixed effects, lagged outcomes, lagged treatments, and other defended lagged state variables.

2 Three diagnostics

For $j \in \mathcal{Z}$, the **one-at-a-time** increment is

$$a_j = \beta(\{j\}) - \beta(\emptyset).$$

It answers how much the coefficient moves when Z_j alone is added to the short model. It holds every other candidate control out, so with correlated controls the values a_j need not add to the joint shift.

The **leave-one-control-out** (LOO) conditional increment is

$$\ell_j = \beta(\mathcal{Z}) - \beta(\mathcal{Z} \setminus \{j\}).$$

It answers how much the coefficient moves when Z_j is deleted from the full model while all other candidates remain. Thus it is a last-in conditional sensitivity, not a share of $\Delta_{\mathcal{Z}}$; across j, these increments need not add to the joint shift.

For an inclusion order σ, write P_j^σ for the controls preceding j. The path-specific sequential increment is

$$c_j^\sigma = \beta(P_j^\sigma \cup \{j\}) - \beta(P_j^\sigma).$$

It telescopes along each path: $\sum_j c_j^\sigma = \Delta_{\mathcal{Z}}$. The **Shapley value** averages this increment over all $q!$ inclusion orders,

$$\phi_j = \frac{1}{q!} \sum_\sigma c_j^\sigma.$$

Consequently, $\sum_j \phi_j = \Delta_{\mathcal{Z}}$. This symmetric allocation averages shared variation across possible orders. It is an accounting convention rather than a causal decomposition, and its factorial enumeration becomes costly as the number of candidate controls grows.

3 Deterministic correlated-control example

The following example is identical in construction to the deterministic fixture used for the multivariate joint-shift derivation. It uses no random-number generation and no paper simulation. All models use the same 240 observations and unweighted OLS.

Table 1: Three control-level diagnostics in the deterministic correlated-control example.

Control	One-at-a-time	Leave-one-out	Shapley
Z1	-0.45785364	-0.32728413	-0.4445487
Z2	0.01218349	-0.08698809	-0.0893821
Z3	-0.87033437	-0.36899951	-0.6716467

The one-at-a-time and LOO columns answer different sensitivity questions, so neither is an additive allocation. The Shapley column is the only one constructed to allocate the joint shift across controls.

3.1 Order dependence

Table 2: Sequential contributions under two valid inclusion orders.

control	contribution (Z1 -> Z2 -> Z3)	contribution (Z3 -> Z2 -> Z1)
Z1	-0.4578536	-0.32728413
Z2	-0.3787244	-0.00795902
Z3	-0.3689995	-0.87033437

The two paths both sum to the same joint shift (-1.20557751835), but the entries assigned to individual controls differ. This is order dependence, not a numerical error.

4 Reproducibility checks

Table 3: Reproducibility checks for the allocation diagnostics.

check	metric	criterion	result
Shapley contributions sum to the joint shift	6.661e-16	error $<$ 1e-10	PASS
Sequential paths telescope to the joint shift	1.776e-15	error $<$ 1e-10	PASS
Shapley allocation is invariant to a permutation of control names	2.220e-16	error $<$ 1e-10	PASS
Sequential control contributions differ across displayed orders	5.013e-01	difference $>$ 1e-4	PASS
One-at-a-time and LOO diagnostics differ for at least one control	5.013e-01	difference $>$ 1e-4	PASS

The name-permutation check permutes the columns and their names together, then realigns estimates by control name. It checks that the Shapley allocation is independent of an arbitrary input-column order.

5 Reporting guidance and limitations

For the paper's linear TSCS/CET use case, report the endpoint shift and LOO increments as transparent conditional sensitivity checks. Report one-at-a-time increments only when the short-model comparison is substantively useful. Reserve Shapley values for an online appendix or replication package: they are a coherent symmetric allocation of a projection shift, but they do not determine a unique causal contribution of correlated controls.

The functions in `scripts/shift_allocation_correlated_controls.R` require a common complete sample and full rank for every requested nested model. They enumerate all $q!$ orders, which is appropriate for a small, prespecified set of candidate controls but not automatically for high-dimensional specifications. They do not cover weighted least squares, non-nested models, regularized estimators, missing-data-induced sample changes, or causal identification of the CET.